

PROJECT TRICORDER

MK I · FIELD OPERATIONS MANUAL

On-device multimodal field instrument: inquiry, vision, image synthesis, neural speech in/out, offline translator, audio scene analysis, avian and flora/fauna classification, artificial horizon with G-meter and GPS speed, Celestrak/SGP4 satellite plot, major-star map, SOL space weather and inner-system orbits, NWS earth forecast, NOAA / flight / civic watch, append-only field log, and optional tricorder-to-tricorder mesh.

Field	Value
Software version	0.6.8 (versionCode 43)
Application ID	com.projecttricorder.app
Platform	Android 8.0+ (API 26), arm64-v8a
Reference hardware	Samsung Galaxy Z Fold3 (Snapdragon 888, 12 GB)
Document date	6 September 2026
Classification	Unclassified — operator manual
Companion artifact	Tricorder-0.6.8-earth-scroll.apk
Supersedes	0.6.4 operations manual (5 September 2026)
Design	Jim Ames, Claude, and Grok · N2NHU Labs · Newburgh, New York 12550
Edition name	RENT-A-HAL Android Edition, implemented as Project Tricorder Mk I

NAVIGATION AND CERTIFICATION WARNING

This instrument is NOT certified by the FAA, NASA, EASA, ICAO, or any civil or military aviation or spaceflight authority. The artificial horizon, heading rose, G-meter, GPS ground-speed tape, satellite look-angles, star map, schematic solar-system chart, and space-weather / HF-propagation displays are uncertified consumer-sensor and public-catalog visualizations. They MUST NEVER be used as primary or backup navigation, attitude, approach, air-data, celestial-navigation, or operational HF circuit references. Do not use Project Tricorder to land, taxi, fly, dock, station-keep, or time a satellite pass for safety-of-flight. Treat horizon, sky, and SOL plots as field curiosities only.

1. Purpose and scope

This is the operator and installer document for Project Tricorder Mk I through software 0.6.8. It supersedes Rev 0.6.4 and adds the SOL SYSTEM face: NOAA Space Weather Prediction Center indices, HamQSL HF/VHF/UHF propagation, a schematic inner-system orbit chart, and a scrollable NWS earth forecast for the GPS fix.

2. What changed since Rev 0.6.4

Build	What landed
0.6.5	O Sol button. Earth-link init fetches NOAA SWPC (Kp, GOES X-ray, scales, 3-day text, D-RAP), HamQSL solarxml, and NWS hourly + period forecast. Schematic Mercury–Saturn orbits. Kokoro speaks a briefing.
0.6.6	HamQSL band names are grouped (80m-40m, 30m-20m, 17m-15m, 12m-10m). Parser now matches those keys so HF rows are no longer dashes.
0.6.7	O cycles three ways: space (orbits + space wx + HF/VHF/UHF) → earth only → off. Button stacks O /

	Sol / off sol wx so the label is not Soloff. UHF aurora is its own line.
0.6.8	Earth pane is a scrollable text panel, not a clipped canvas. Six hours plus today and the next two days, including NWS detailed lines.

Features already documented in 0.6.4 and still current: I–IV modes, translator wheels and AUTO, horizon / G-meter / GPS speed, Sat/Stars SGP4 plot, watches, field book, Camp link, first-run pack fetch, Help / Credits / Warning / on-device Manual.

3. Face controls (0.6.8)

Control	Function
I Inquiry	Gemma chat; Imagine keywords; COMPUTER easter egg; exact HELP opens the briefing
II Sensor	Camera + SmoIVLM + flora/fauna on Hold
III Sentry	Motion then caption; Hold after each hit unless Continuous is on
IV Audio	YAMNet duty cycle, BirdNET on bird-like hits, watches if checked
T Translator	Offline speech-to-speech; SOURCE / TARGET wheels; AUTO swap
H Horizon	Green = tape on (idle + Audio). Red = off so the CRT can show the log
S Sat/Stars	Cycles satellites → stars → off. Polar live plot from Celestrak + SGP4
O Sol	Cycles space → earth weather → off. See §7.
L / E / V / P	Log review slider, officer note, video log, play current entry

4. Architecture additions

SolStore.kt runs on the same earth-link init thread as TLE fetch. Snapshots write to files/sol/snapshot.json. Public feeds, no API keys. User-Agent identifies the instrument (NWS requires one). satellite.js remains in assets for S; SOL orbits are mean-element clock-face math, not SGP4 and not JPL Horizons.

Feed	URL family	Used for
NOAA SWPC JSON/text	services.swpc.noaa.gov	Kp, GOES X-ray, NOAA scales, 3-day forecast, D-RAP HF absorption
HamQSL / NONBH XML	hamqsl.com/solarxml.php	SFI, SSN, A/K, X-ray class, grouped HF day/night, VHF aurora / Es
NWS API	api.weather.gov/points/{lat},{lon}	Hourly + 12-hour period forecast at the GPS fix
Celestrak GP/TLE	celestrak.org (already in 0.6.4)	Sat/Stars only — not the SOL orbit sketch

5. Theory — SOL path

5.1 Init

If 8.8.8.8 answers, the APK fetches SWPC + HamQSL + NWS after the TLE packs. Status lines appear on the CRT. Fetch SOL wx on the SOL bar refreshes the snapshot without a reboot. A snapshot taken before 0.6.6 lacks grouped band keys — tap Fetch SOL wx once after upgrading.

5.2 Space page

First O: schematic orbits of Mercury through Saturn from mean longitude vs days since J2000. Right-hand column is SFI, SSN, A, Kp, Bz, solar wind, X-ray class, HF groups day/night, VHF aurora (northern hemisphere

from HamQSL phenomenon tags), then a dedicated UHF aurora line. UHF is not a HamQSL product; the line is derived from Kp (quiet / line of sight, possible scintillation at $Kp \geq 5$, scintillation likely at $Kp \geq 7$). Kokoro speaks the space briefing.

5.3 Earth page

Second O: the orbit canvas hides. A scrollable NWS pane shows city/office, the next six hourly periods, then Today + next two days (six 12-hour periods) with short forecast and detail text. Drag to read past the fold. Kokoro speaks hours + outlook. Third O turns the feature off. The button caption under Sol reads off, sol, or wx.

5.4 Limits

Orbits are a clock face, not an ephemeris. HF conditions are a model, not a circuit guarantee. NWS coverage is CONUS and affiliated offices; a fix outside NWS grids yields an empty earth pane. D-RAP text is stored in the snapshot but the face uses HamQSL for the band table.

6. Operations still current from 0.6.4

Inquiry, Sensor, Sentry, Audio, translator, Imagine sliders, horizon (portrait G-meter + GPS SPD), Sat/Stars, watches, Camp link, field book, and first-run BirdNET / flora / translator-pair fetch are unchanged. See Rev 0.6.4 §§4–8 for those paths. The certification warning now applies to G, speed, satellites, stars, and now SOL.

7. Guide — SOL SYSTEM

Allow Location so NWS and the observer site are real. Boot on Wi-Fi and wait for “fetching SOL.”

Press O once. Confirm SFI / Kp / X-ray and that HF rows are Fair/Good/Poor, not dashes. Press O again. Scroll the earth pane through tonight and the next two days. Press O a third time to clear the CRT.

If HF is still dashes, Fetch SOL wx. If earth is empty, walk outdoors for a GPS fix and fetch again.

8. Capabilities matrix (additions)

Capability	Where	Network	Limit
SOL space wx	APK SolStore	Once per init / Fetch	SWPC public JSON
HF / VHF table	HamQSL XML	Once per init / Fetch	Grouped bands; UHF derived
Earth forecast	NWS API	Once per init / Fetch	CONUS-centric; scroll pane
SOL orbits	JS mean elements	None	Schematic, not JPL

9. Troubleshooting additions

Symptom	Likely cause	Action
HF rows are dashes	Old snapshot keys (pre-0.6.6) or failed HamQSL	Fetch SOL wx on Wi-Fi
O does not cycle / always shows both	Build older than 0.6.7	Install 0.6.8
Button reads Soloff	Tiny label inline on 0.6.6	Install 0.6.7+
Earth stops at Today + next 2 days	Canvas clip on 0.6.7	Install 0.6.8 and scroll
Earth pane empty	No GPS or outside NWS	Location on, Fetch SOL wx

SOL silent	Kokoro :5000 down	Start Termux tts; text still prints
------------	-------------------	-------------------------------------

10. Document control

Rev	Date	Notes
0.5.4	4 Sep 2026	Translator, horizon, NOAA/OpenSky, FAA warning
0.6.4	5 Sep 2026	G-meter, GPS speed, Sat/Stars, civic watch, first-run fetch
0.6.8	6 Sep 2026	SOL SYSTEM three-step cycle, SWPC + HamQSL + NWS, scrollable earth pane

Prepared for operator use. N2NHU Labs, Newburgh, New York 12550.